

QT Interval Prolongation

Characteristics: The QT interval in the cardiac cycle represents depolarization (contraction) and repolarization (relaxation) of ventricles (these interval times are corrected to account for variations in heart rate, specified as "QTc"). Normal QTc intervals are generally defined as <460 msec for women and <450 msec for men. QT prolongation can lead to serious arrhythmias, including torsades de pointes (TdP), and sudden death. While the link between QT and TdP is not clear, QTc above 500 msec is a significant risk factor for TdP.

Meds That Cause It: Typical antipsychotics (especially thioridazine, high doses of chlorpromazine, and intravenous haloperidol). Of atypical antipsychotics, ziprasidone has greatest risk of prolonging QT, though to a lower degree. Of SSRIs, higher-dose citalopram (>40 mg/day) most likely. Of TCAs, amitriptyline and maprotiline. Methadone. In overdose, a large number of medications can prolong QT.

Mechanism: The mechanism depends on the offending drug. For example, TCAs prolong the QT through blockade of sodium and calcium channels, while citalopram and chlorpromazine do so through blockade of potassium channels.

Management:

- Prevent: Identify risk factors (see below), get baseline electrolytes, ECG, and monitor periodically in patients at risk.
- With prolonged QTc (>500 msec), obtain a cardiologist consult.
- In patients with a borderline QTc, switch to a less risky agent.
 - Lower-risk antipsychotics: Aripiprazole has minimal risk; asenapine, lurasidone, olanzapine, and quetiapine are also good options.
 - Lower-risk antidepressants: All antidepressants at usual therapeutic doses are relatively safe (avoid citalopram >40 mg/day). However, sertraline may be the best choice because it is the most studied in cardiac patients and has few drug interactions.

Clinical Pearls:

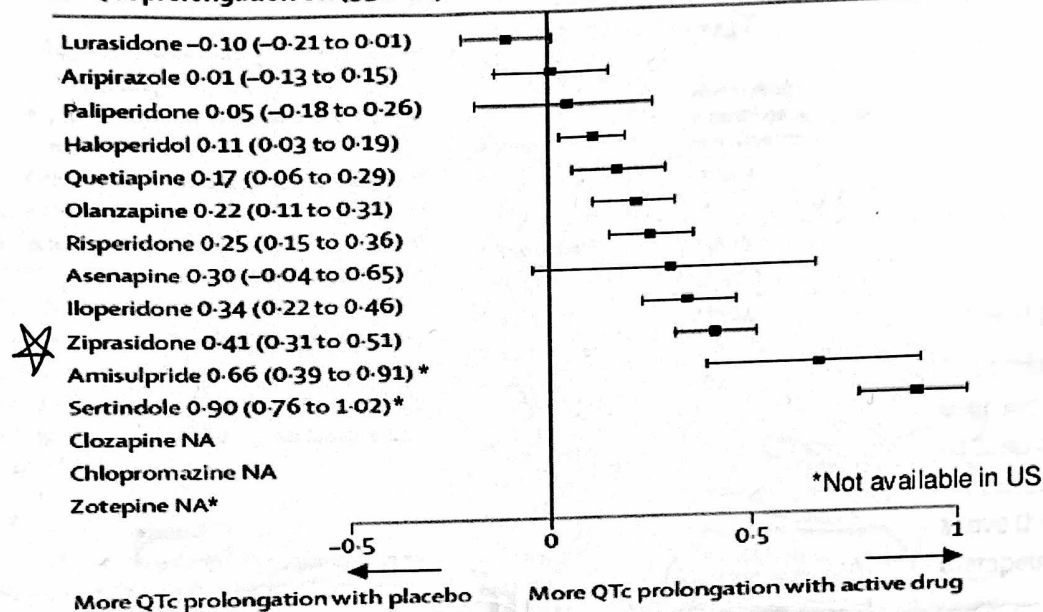
- Risk factors: Congenital (long QT syndrome), female gender, older age, electrolyte abnormalities (low potassium, calcium, magnesium), and other conditions such as hepatic dysfunction, diabetes, hypothyroidism, AIDS, and hypertension.
- Heart conditions that increase a patient's risk are bradycardia, left ventricular dysfunction, heart failure, mitral valve prolapse, and myocardial infarction.
- Caution in situations that may affect electrolytes, including eating disorders, diuretic use, hypoglycemia, renal dysfunction, and pituitary insufficiency.
- Other common medications that are not related to psychiatry can also be culprits, including antiarrhythmics such as sotalol, amiodarone, and quinidine; macrolide antibiotics such as azithromycin; quinolone antibiotics such as levofloxacin; some antifungals; antimalarials; and other medications such as tamoxifen.
- Overdose with psychotropic medication, even "low risk" agents, can lead to QT prolongation.

Fun Fact:

- *Torsades de pointes* translates to "twisting of the points," illustrating the chaotic nature of the heart rhythm that can lead to sudden death.

QTc prolongation

E QTc prolongation OR (95% CrI)



Comparative efficacy and tolerability of 15 antipsychotic drugs in schizophrenia: a multiple-treatments meta-analysis. *Lancet* 2013; 382: 951-62

Which psychotropics carry the greatest risk of QTc prolongation?

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Mrs. A, age 68, has a 40-year history of schizoaffective disorder with comorbid anxiety disorder not otherwise specified, type 2 diabetes mellitus, and hypertension. She takes furosemide, 40 mg/d, lisinopril, 20 mg/d, and metformin, 2,000 mg/d, for hypertension and diabetes; lorazepam, 1.5 mg/d, and paroxetine, 40 mg/d, for anxiety; and quetiapine extended release, 800 mg/d, for psychotic features and mood dysregulation with schizoaffective disorder. Mrs. A's husband died 5 years ago and she lives alone in a senior care facility. Mrs. A uses a weekly pill reminder box because her residential facility does not monitor medication adherence. She sees her psychiatrist once a month and her primary care provider every 3 months. She has no history of illicit drug, alcohol, or tobacco use.

Two weeks ago, Mrs. A was found leaning against the wall in a hallway, complaining of dizziness and disorientation, and unable to find her way back to her apartment. In the emergency department, her serum potassium is low (3.0 mEq/L; normal range: 3.5 to 5.0), fasting glucose is elevated (110 mg/dL; range: 65 to 99), and ECG reveals a prolonged QTc interval of 530 milliseconds. Before this episode, Mrs. A had been medically stable without mood or psychotic symptoms, although her daughter reported medication self-administration was becoming difficult.

Exposure to psychotropics carries a risk of QTc prolongation. The QT interval is

an ECG measure of ventricular depolarization and repolarization. The QTc designation indicates a correction for heart rate with increasing heart rate correlating with a shorter QT interval. Readings of 440 milliseconds are considered normal.¹ QTc prolongation is defined as >450 milliseconds for men and >470 milliseconds for women.² An increase in the QT interval is a predictor of serious cardiac events.³

Antidepressants and antipsychotics have been associated with QTc prolongation. When identifying agents that could disrupt cardiac conduction, clinicians need to consider whether the drug's molecular structure, receptor affinity, or pharmacologic effects are most critical.² Although these may be important, patient-specific variables that increase the risk of QTc prolongation may have greater impact. These include:

- age >65
- female sex

Practice Points

- Screen patients for risk factors for prolonged QTc interval, such as congenital long QT syndrome, family history of cardiac conduction abnormalities, and previous occurrences of medication-mediated QTc prolongation.
- Obtain baseline and steady state ECG when initiating high-risk agents, particularly when administering combination therapy.
- Use the lowest effective dose of antidepressants and antipsychotics and monitor symptoms closely.

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Clinical Point

QTc prolongation has been associated with long-term ziprasidone use more often than with risperidone, olanzapine, or quetiapine

Related Resources

- De Hert M, Detraux J, van Winkel R, et al. Metabolic and cardiovascular adverse effects associated with antipsychotic drugs. *Nat Rev Endocrinol*. 2011;8(2):114-126.
- Vieweg WV, Wood MA, Fernandez A, et al. Proarrhythmic risk with antipsychotic and antidepressant drugs: implications in the elderly. *Drugs Aging*. 2009;26(12):997-1012.
- Sandson NB, Armstrong SC, Cozza KL. An overview of psychotropic drug-drug interactions. *Psychosomatics*. 2005;46(5):464-494.

Drug Brand Names

Amantadine • Symmetrel	Lurasidone • Latuda
Amitriptyline • Elavil	Maprotiline • Ludiomil
Aripiprazole • Abilify	Mesoridazine • Serenitil
Asenapine • Saphris	Metformin • Glucophage
Chlorpromazine • Thorazine	Nortriptyline • Pamelor
Citalopram • Celexa	Olanzapine • Zyprexa
Clozapine • Clozaril	Paliperidone • Invega
Desipramine • Norpramin	Paroxetine • Paxil
Fluoxetine • Prozac	Pimozide • Orap
Furosemide • Lasix	Quetiapine • Seroquel
Haloperidol • Haldol	Risperidone • Risperdal
Hydroxyzine • Atarax,	Tamoxifen • Nolvadex,
Vistaril	Soltamox
Iloperidone • Fanapt	Thioridazine • Mellaril
Lisinopril • Prinivil, Zestril	Venlafaxine • Effexor
Lorazepam • Ativan	Ziprasidone • Geodon

Disclosures

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mesoridazine are well-documented and thioridazine-mediated QTc prolongation increases are dose-dependent.^{4,18} ECG monitoring is recommended with IV haloperidol, which is used for delirium in adults.²⁰ QTc prolongation has been associated with long-term ziprasidone use more often than with risperidone, olanzapine, or quetiapine.¹⁹ Ziprasidone prolongs the QTc interval an average of 20 milliseconds,²¹ which could represent a clinically significant change. QTc prolongation for iloperidone is comparable to ziprasidone and haloperidol.²² There is some evidence that aripiprazole may shorten, rather than prolong, the QTc interval.^{4,17}

Cardiovascular adverse effects associated with clozapine—including QTc prolongation—are dose-dependent.³ Olanzapine prolongs QTc interval, although the mean

change is less than with other agents unless other variables were present, such as:

- concomitant use of medications that may prolong QTc interval (ie, amantadine, hydroxyzine, or tamoxifen²)
- preexisting cardiovascular conduction disorders
- higher doses (>40 mg/d).^{3,23}

In 17 case reports of cardiac changes associated with quetiapine use, doses ranged from 100 mg/d²⁴ to an overdose of 36 g/d.²⁵ Only 1 patient death was reported secondary to overdose and preexisting dysrhythmia and hypertension.²⁶ QTc prolongation associated with risperidone was minor¹ based on oral doses in the normal therapeutic range and incidences of overdose.¹⁰ Paliperidone²⁷ and lurasidone²⁸ are associated with clinically insignificant QTc prolongation. Changes in QTc interval were positively correlated with asenapine dose, although at the highest dose of 40 mg/d, the increase was <5 milliseconds.²⁹

Mrs. A presents with a number of risk factors for QTc prolongation, including older age, female sex, and psychiatric and medical comorbidities that require medication. A pill count revealed that she was taking more than the prescribed daily doses of her medications. During the interview, Mrs. A said that if she missed her medication time, she would take them when she remembered. If she could not remember if she took her pills, she would take them again. Her physicians will explore strategies to increase medication adherence.

References

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